FOR TWO IMMERSIONS OVER A SMOOTH MORPHISM THE CONORMAL SEQUENCE IS SHORT EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 7454-7472 and proof span 7474-7485 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains an xymatrix triangle of immersions; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. The proof is the author's algebraic reformulation and citation, with no step added.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The conormal sequence relating the conormal sheaves of two immersions in a commutative triangle, which is exact at the middle and right terms, cited from the same chapter. Cited at proof lines 7466-7472. Supplies the sequence whose exactness at the left term has to be proved.
- The algebraic statement that for ring maps with a surjection and a smooth second map the corresponding sequence of conormal and differential modules is exact, cited from the algebra chapter. Cited at proof lines 7474-7485. Gives the affine computation yielding the missing exactness.

Source reference. Original lemma 065V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7914–7914. Same-chapter reference used in the retained author text at source line(s) 6847, 7482. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 06A9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8068–8068. Same-chapter reference used in the retained author text at source line(s) 7484. It is supplied only as a navigation pointer

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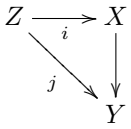
Source reference. Original lemma 067L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7975–7975. Same-chapter reference used in the retained author text at source line(s) 7470. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let*



be a commutative diagram of schemes where i and j are immersions and $X \to Y$ is smooth. Then the canonical exact sequence

$$0 \to \mathcal{C}_{Z/Y} \to \mathcal{C}_{Z/X} \to i^*\Omega_{X/Y} \to 0$$

of Lemma 067L is exact.

Proof. The algebraic version of this lemma is the following: Given ring maps $A \to B \to C$ with $A \to C$ surjective and $A \to B$ smooth, then the sequence

$$0 \to I/I^2 \to J/J^2 \to C \otimes_B \Omega_{B/A} \to 0$$

of Algebra, Lemma 065V is exact. This is Algebra, Lemma 06A9. □